22 05 00 - Common Work Results for Plumbing

Following is a brief description of the features which should be incorporated in the specifications for various plumbing materials and equipment. Where a specific product is stated with model number, that product should be used as the basis of design and listed as the first named in the specifications. Manufacturers listed without model number are intended to give the designer input on manufacturers whose products have been found to be acceptable.

General Quality Level:

Generally, plumbing material and equipment selected should be institutional grade. Long life and simple low cost maintenance are critical attributes for nearly all CAL POLY projects. Plumbing fixtures in public areas should also be selected to endure rough use and daily janitorial cleaning.

General-Duty Valves Requirements:

- Isolation Valves: Provide sufficient number of valves for ease of service, and to reduce inconvenience to users due to outages and draining of systems for minor repairs.
- Ball valve: Provide with stainless steel ball and stem from ½”-1.5” and from 2” up either bronze gate valve or epoxy coated resilient wedge iron body gate valve.
- Relief valves: Daylight to a conspicuous location. Plumb to sewer.

Identification for Plumbing Piping and Equipment Requirements:

- Identification Charts: Location should be reviewed and approved by Project Manager in consultation with Facility Services.
- Piping Service: Identification Tags: Provide with abbreviated legend on 1st line and pipe size on 2nd line. Locate to be visible from exposed points of observation. Where 2 or more pipes run parallel, place printed legend and other markers in same relative location.
- Valves Service Identification Tags: Provide with abbreviated legend on 1st line and valve service chart number on 2nd line. Identification Charts: Provide two (2) satin finished extruded aluminum frames with rigid clear plastic glazing; 8-1/2 x 11 inches, minimum for each chart. In addition, provide electronic copy of each chart.

Keys for Cabinets and Padlocks:

- Cabinets and Equipment: Provide 2 keys per panel. Coordinate with Section 08 06 05 – Key Schedule.
- Padlocks: Coordinate with Section 08 06 05 – Key Schedule.
- Closeout Submittal: Provide panel keys separated and labeled. Provide location, room number, quantity, manufacturer name and model numbers of keys, and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

Testing

- All new piping shall be tested prior to tie in to existing systems.
- Do not test against existing valves when connecting into an existing system. Provide a slip blind at the valve flange or other suitable isolation.
- Test gauges shall have 3” minimum dial, with oil fill and gauge cock. Gauge calibration shall be verified to the satisfaction of the University’s Inspector prior to commencing testing.
- Domestic and Industrial Hot & Cold Water Piping
- 150 PSIG test pressure for a period of 4 hours using a test medium of water.
- 200 PSIG test pressure for a period of 4 hours using a test medium of water for portions of the system which serve both the fire suppression system and the domestic water .
- Drain, Waste & Vent Piping Including Lab Waste: 10 feet of head for a period of 1 hour using a test medium of Water.
- Natural Gas Piping: 50 PSIG test pressure for a period of 4 hours using a test medium of air.
- Compressed Air Piping: 150 PSIG test pressure for a period of 4 hours using a test medium of air.
- Pure Water Piping Systems: 100 PSIG test pressure for a period of 4 hours using a test medium of purified water.22 05 19 Meters and Gages for Plumbing Piping
    - Water meters for domestic water should be nutating disk type sufficiently accurate to record the lowest expected flow (typically the flush of a single low flush toilet). Provide compound metering if necessary to accommodate a large range of flows.
    - Provide a permanently piped full size meter bypass on all domestic water services 3” and larger. Valving should be arranged so that the meter can be removed without a disruption of water service.
    - All water meters should record in cubic feet.
    - Meters for irrigation may be either nutating disk or turbine type depending on expected flow.

22 06 00 - Schedule for Plumbing

Boiler / Chiller Room Plumbing

Provide floor drains / sinks for relief valve drains and to allow for drain down of closed loop systems. Coordinate locations with equipment layout so that piped equipment drains will not be a trip hazard in service aisles. Select floor drains/sinks to contain spill over and splashing due to equipment drain discharge.

Provide make up water through a RP backflow preventer for each closed loop system.

Plaster / Sand Use.

Rooms where plaster or other sediment material is used shall be equipped with a dedicated above grade sand trap on the drain outlet of each sink. Typical applications include art studios using plaster, anthropology labs using plaster for bone castings, etc.

Dark Rooms
Provide RP back flow preventers for hot and cold water to dark room sinks.

Fume Hoods
For each plumbing fume hood plumbing service, provide a manual isolation valve and union at the piping drop above each fume hood so that the hood can be easily removed with minimal disruption of building piping.

Commercial Kitchens
Per City of San Luis Obispo Waste Water Discharge Ordinance, a large outdoor grease trap is typically required. Position grease trap to allow for easy access for pumping and minimal nuisance odors at occupied locations while pumping.

22 06 10 - Schedule for Plumbing Piping and Pumps
Piping: Standard: U.S.-manufactured.

22 10 00 - Plumbing Piping
All underground plumbing utility piping shall have a minimum of 12” sand encasement.

All underground plumbing utility piping shall have a minimum depth of 24” unless otherwise approved by the administrative authority after all existing utilities have been potholed by the contractor.

Design Documentation
Programming; For wet laboratories, studios, dark rooms, commercial kitchens, laundries, and other special use rooms requiring plumbing utilities to accommodate special functions, the Design Professional shall gather and document all information required to confirm Plumbing requirements as part of room programming. This information shall be included with the Design Development Submittal. Obtain and document the following information for each building room where special use plumbing services will be required.

- Room name
- Person requesting the plumbing services, Interviewer, Date
- Room use
- A list of required plumbing services. Include qualitative and quantitative data if available.
- Functions / processes which the required plumbing services are intended to accommodate. Provide sufficient information to ascertain the need for special plumbing provisions such as back flow prevention, pressure regulation, drain sediment traps, etc.

- A list of all equipment / apparatus which will require plumbing services. Include equipment manufacturer's data (connection size, use rate, etc.) if available.
- The above programming information may be omitted for rest rooms and residential kitchens where all required design data is covered by applicable codes.

Calculations:

The Design Professional shall include final plumbing calculations with the 50% Working Drawing submittal. These calculations shall include the following:

- The room programming documents covered above.
- All assumptions clearly stated. In particular, document assumptions that had to be made in order to proceed with the design in the case of insufficient data being provided by the users. The intent here is to flag assumptions which the University should verify as correct.
- Applicable code requirements
- Tabulated design criteria for each plumbing service by system including:
    - System
    - Location of Service (Room number)
    - Required flow quantity for each location including: water use fixture units, drainage fixture units, GPM flow rates for specialized equipment, natural gas load by both BTU/HR and CFH, CFH for air, vacuum & specialized gasses.
- Calculated data as required to select each piece of plumbing equipment including:
    - Hot water heater demand and storage capacity.
    - Sump pump sizing data.
    - Lift station sizing data.
    - Circulation pump sizing data.
- Pipe sizing data including: combined flow for each pipe run, developed length for each pipe run, selected pipe size based on stated criteria (sizing chart, pressure drop, fluid velocity, etc. This data may be either in tabulated form or rough pipe schematic.
- Rainwater leader calculations

Plumbing Working Drawings

General: Plumbing Working Drawings shall be sufficiently complete to assure high quality, fully functional plumbing systems, no contractual ambiguity, and also shall serve as a long term record of the building's plumbing systems. The following guidelines apply to all Plumbing

- No work shall be called out in a manner which is not contractually enforceable. Plumbing quality level shall be fully identified to assure fair competitive bidding as well as a quality installation.
- Design / build format for the plumbing section of the work as is sometimes used in residential plumbing construction shall not be used on Cal Poly projects. An exception to this guideline shall be the case of the entire building project being design / build format when approved by the University's Representative.

- All points of connection to existing piping systems shall be identified.
- All piping shall be sized.
- The routing of all piping shall be indicated. The indicated routing shall be verified to be feasible within the furring spaces indicated in the architectural drawings.
- All equipment and fixtures shall be located. Maintenance access space as recommended by the manufacturer shall be indicated on the drawings.
- All piping penetrations of the building shell shall be indicated.
- Concrete core drills of existing structures shall be identified.

Provide a Plumbing Legend covering all symbols and abbreviations used in the plumbing drawings in order to have a fully enforceable contract.

Provide a Plumbing Schedule covering factory assembled plumbing equipment and fixtures. The intent should be the easy determination of the plumbing equipment included in the project to encourage competition among comparable product suppliers. Note that when a product manufacturer's name and model are called out on the schedule to serve as the basis of design, the specifications must be also be coordinated so that the same manufacturer is the first specified. See Section 01630; Product Options and Substitutions of Cal Poly's standard Division 1 for further information.

Provide Plumbing Floor Plans to scale for each level where plumbing services are to be provided. Provide enlarged partial plans for congested areas such as public rest rooms where the normal scale plans will not adequately depict the work.

Sections to scale should be provided when the vertical arrangement of equipment and piping relative to other building components is critical for proper function or adequate maintenance clearance, as well as to assure the feasibility of pipe routing through tight spaces.

Plumbing Riser Diagrams: Plumbing riser diagrams shall be provided for multi-story buildings whenever the relative configuration of piping components cannot be depicted in the plan view alone without ambiguity. As a minimum, riser diagrams shall be provided in the Plumbing Working Drawings for the following systems: all drain waste & vent systems including lab waste, all hot water systems with circulation, all pure water systems, and any other system which includes forced circulation by pumping.

Plumbing Details: Plumbing details shall be provided as necessary to contractually assure a given level of quality. As a minimum, details shall be provided to cover all pipe penetrations of roofs, equipment supports, piping supports on roofs, backflow preventers, equipment installations where the relative position of plumbing appurtenances such as valves, unions, thermometer, drains, etc. are important to function, ease of service, and future replacement, pipe crossings of building expansion joints, pressure reducing assemblies, backflow prevention assemblies, roof drainage assemblies, floor drain assemblies, all pipe penetrations of exterior walls above and below grade.

Piping Diagrams (Schematics) shall be provided whenever the relative arrangements of plumbing components and equipment cannot be clearly depicted in the plans. As a minimum, piping diagrams shall be provided for in the Plumbing Working Drawings for water heating systems with circulation, pure water systems, and lift stations.

22 11 00 - Facility Water Distribution

All back flow preventers will be tested by a certified tester prior to the system by put in service.

Cal Poly owns and maintains the underground water distribution system throughout the campus.

Provide a domestic water pressure reducing valve and meter at each new building.

Provide a separate water meter and RP backflow preventer for irrigation at each new building.

Fire water should not be metered or reduced in pressure. Back flow prevention is not required for fire water.

DCDA assemblies are required by Cal Poly and the County of San Luis Obispo.

For new buildings to be connected to the campus water system, the anticipated additional water demand should be identified early during preliminary planning. This water demand should be submitted to the Principal Engineer for CAL POLY Facilities Planning and Capital Projects.

Improvements to the campus system may be required to accommodate the additional demand. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

Design Conditions:

Water System Pressure: The water pressure assumed for system pipe sizing and design shall be the lower of the following:

- 60 PSI
- The actual system pressure.

Since the water pressure in the campus water mains varies throughout the campus depending on elevation within four water pressure zones, the Plumbing Designer should request the water pressure from the University's Representative for each project. The University's Representative will provide the known pressure at a given point (usually a fire hydrant). The Plumbing Designer will be responsible for determining the pressure at the point of use and should take into account the change in elevation between the known reference point and the point of use.

22 11 16 - Domestic Water Piping

Acceptable Piping Materials

- Above Grade: Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder for joining 2" and smaller pipe. 15% silver brazing conforming to AWS classification BCuP-5 for joining 2-1/2" and larger piping.
- Below Grade: Type K hard drawn or soft copper tubing, with wrought copper sweat fittings, all connections brazed with 15% silver brazing conforming to AWS classification BCuP-5 , wrap pipe with 1 layer of 10 mil tape. Soft copper tubing with radius bends shall be used to minimize below grade fittings. (Note: Pipe layouts which place water piping under the building slab should be avoided whenever other alternatives are feasible.)
- Below grade, beyond building footing, 3" and smaller piping, alternate material: Schedule 80 PVC pipe with socket fittings and solvent cement joints or HDPE.
- Below grade, beyond building footing, 4" and larger: Cement-mortar lined Class 150 ductile iron pipe with bell and spigot joints and cast iron fittings. Specify restraining method: thrust blocks or restrained joints. Restrained joints should be called out for pipe configurations where thrust blocking will be difficult.
- Below grade, beyond building footing, 4" and larger, alternate material: Class 200 AWWA C-900 PVC with bell and spigot joints and cast iron fittings. Specify restraining method: thrust blocks or restrained joints. Restrained joints should be called out for pipe configurations where thrust blocking will be difficult.

Water Pressure Reducing Valves

Provide a pressure reducing valve (PRV) which will limit the supply pressure to 80 PSI at the domestic water entry point to each building. This PRV shall be provided on water services with pressure below 80 PSI as well as above in order to provide additional protection in the event of a loss of pressure control in the campus water mains.

For low system pressures where a PRV may cause an undesirable pressure loss an exception to this requirement may be granted upon obtaining approval from the University's Representative.

Provide compound PRV's where required to accommodate a large range of flows.

Fire water should not be reduced in pressure. Separate fire water from domestic water ahead of the PRV.

Valving:

Water piping systems shall be provided with manual isolation valves at the following locations:

- At branch connections to underground system mains.
- At all building points of entry.
- On both sides of piped in devices which may need to be removed for servicing including water meters, back flow preventers, pressure reducing valves, strainers, pumps, etc. The devices shall be removable without draining down the building system.
- Exception: On small branch lines (1 inch and less) where the quantity of building water is small, the valve on the downstream side of the device may be omitted.
- At each floor where branch piping connects to a riser.

Domestic / Industrial Hot Water

For water conservation, all systems shall be designed to provide near immediate hot water at fixtures by using either a hot water circulation pump or electric heat tape. An exception to this is small systems with a short distance (50 feet or less) between the water heater and the fixture. Verify with the University Representative whether heat tape or circulation will be used on a project by project basis.

For systems with a hot water circulation, verify that the pump is not over selected and that the pipe velocities due to circulation will be reasonably low. (Copper fitting failure due to high velocity scouring has been a problem). Grunfos Alpha Series Pumps are preferred.

Hot water mixing valves where required shall be Powers Intellistation. Intellistation Jr., or equal meeting the criteria below:

- Configurable on location. Does not require factory pre-programming or special software and laptop
- Controls water temperature to +/- 2°F during periods of low/zero demand
- 3.5° full-color, user-selectable touch screen display
- User programmable high-temperature sanitization mode
- In case of power or cold water failure, valve flows full cold for safety with manual override feature to set mixed outlet temperature
- Settings can be adjusted/monitored at the controller or remotely through BAS (Building Automation System)
- Displays pressure, temperature and flow/BTU data

- Pass code protected for security
- User programmable high temperature alarm

Domestic hot water heaters shall be gas fired, to minimize campus electrical demand as well as energy use cost.

Exceptions:

- Domestic hot water in buildings served by the campus central heating water system should be provided from a storage tank equipped with a double wall heat exchanger. (An exception to this may be allowed in the case of buildings with very low anticipated domestic hot water usage).
- Lavatories with small usage which are remotely located from a central system may be provided with hot water from a small electric storage type water heater with a maximum electrical demand of 1.5 kW. This exception shall not apply to shower usage. Instantaneous electric water heaters are not permitted due to high electrical demand.

Provide separate heating sources for domestic hot water and space heating water except for buildings served by the campus central heating water system.

Water heaters shall be installed in a sheet metal drain pan (smitty pan) with drain outlet piped to a safe location on the building exterior or floor drain.

Exception; Water heaters installed on a concrete floor slab where leakage would not damage the floor or building. Provide a floor area drain in the vicinity of the water heater.

All hot water piping shall be insulated.

Water systems shall be designed to prevent water hammer. Provide properly sized shock arrestors adjacent to all quickly closing valves including toilet flush valves, washing machines, dish washers, and solenoid valves. Provide a ball valve below each shock arrestor to allow for removal without system drain down. Provide access doors for all shock arrestors in concealed locations. Shock arrestor locations and sizes should be positively identified within the contract documents.

Disinfection
All potable water systems shall be disinfected and analyzed for bacteriological content. (Note: On Cal Poly projects, fire protection systems should also be disinfected, since they are installed without approved backflow protection) No fire system shall be installed without back flow protection.

Bacteriological analysis shall be completed by a third party laboratory approved by the Cal Poly Office of Environmental Health & Safety (EH&S). The laboratory shall be submitted via the University's Representative for approval by EH&S a minimum of 72-hours prior to conducting disinfection. Analysis results shall be submitted via the University Representative to EH&S to certify compliance with the specifications. Disinfection procedure shall be repeated should EH&S indicate that compliance with the applicable regulations has not been achieved. Plumbing shop representative will need to be given a copy of the results before the system can be put into service.

Provide an industrial water system for non-potable uses at all wet laboratories, faucets with serrated tip for hose connections, dark rooms, and all other locations using toxic or hazardous materials and requiring water service for uses other than drinking. Backflow protection shall be by a reduced pressure principal type back flow preventer.

22 13 00 - Facility Sanitary Sewerage
Cal Poly owns and maintains the sanitary sewer system throughout the campus.

For new buildings to be connected to the sanitary sewer system, the anticipated additional sewer load should be identified early during preliminary planning. This load should be submitted to the Principal Engineer for Cal Poly Facilities Planning and Capital Projects. Improvements to the campus system may be required to accommodate the additional load. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

Systems discharging to the campus sewer system shall be in compliance with the requirements of the City of San Luis Obispo Waste Water Treatment Plant. Mechanical systems discharging unusual wastes may require special provisions or may not be allowed. Triggers include; high or low pH, oil, grease, chemical contamination, biological contamination, and rain water to the sewer. Mechanical systems typically impacted include: commercial kitchen waste, elevator sump pump discharge, lab process waste, and water purification system waste. The Design Professional shall be responsible for confirming impacts to the project due to the City Waste Water Discharge Ordinance. Alternatives for compliance shall be discussed with the University’s Representative. Negotiations involving specific project issues shall be channeled through the Cal Polly Office of Environmental Health & Safety (Director).

For wet laboratories, the method which will be used for dealing with contaminated liquid wastes should be determined early in the design process in consultation with the University's Representative and Cal Poly Environmental Health and Safety (EH&S) . Contaminated liquid waste from laboratories are not allowed to be poured directly into drainage systems discharging to the campus sewers. EH&S should be consulted for procedures for dealing with liquid contaminated wastes. Contaminated wastes are typically containerized and held in the laboratory for disposal by EH&S as hazardous waste. Lab waste piping systems are provided in wet laboratories only for disposal of non-hazardous liquids and as a precaution in case of an accidental spill.

Soil and Waste Piping within a building and outside within five feet (5') of the foundation, except where indicated otherwise, shall be No-Hub cast iron pipe and fittings, asphalt coated, free from defects, and shall conform to the requirements of CISPI Standard 301 ASTM A-888 or ASTM A-74 and manufactured by AB & I, Charlotte or Tyler. Fittings shall be up with Stainless Steel, Heavy-Duty No-Hub Couplings and shall be in compliance with ASTM C-1540 and ASTM C-564 Standards, except all above ground vent piping joints may be made up with Standard-Duty No-Hub Coupling in compliance with CISPI –310, and ASTM C-564 Standards. For fats, oils and grease waste, Blucher Pipe systems are preferred.

Complete soil/waste drainage piping will be provided to each domestic plumbing fixture and will discharge by gravity. Sanitary drainage ventilation piping will be provided to each domestic plumbing fixture or trap and will terminate at various locations on the roof.

Horizontal sanitary, waste and vent piping will be designed for a uniform grade of 2% (¼" per foot).

HVAC condensate drainage piping (CD) will be provided to each separate HVAC unit, such piping will drain to an indirect waste connection to the sanitary soil/waste system via either tailpiece connection at the nearest lavatory or sink or indirectly to a floor sink.

Floor drains will be provided in laundry rooms and all toilet rooms. Drains will not be located under the partitions and will be provided with clearance for service work.

Underground cast iron pipe will be wrapped with 8 mil polypropylene pipe wrap to protect against low pH soil on campus.

All floor drains will be served by a trap primer. The trap primers will be flush-o-meter valve flushing tube or lavatory p-trap type. (No mechanical type trap primers will be provided).

Wall clean-outs at the end of runs for main lines should be provided, where possible above rim height of highest fixture on that line of all floors.

Shower drains will be fitted with removable hair screens within floor drain for ease of maintenance.

Sink drains will be cleanable without disassembly of the associated trap with cleanout above rim height of sink on all floors.

Routing drainage piping overhead of electronic equipment, telecomm equipment and electrical equipment will be avoided.

Cleanouts will be the same nominal size as the pipe they serve; where they occur in piping eight inches and larger, six inches size will be used. All cleanouts will be accessible.

Cleanouts will be provided on all floors at the following locations:

- Horizontal offsets.
- End of soil, waste, water or storm drains more than five feet in length.
- Maximum of 50 foot intervals of horizontal runs within the building.
- Base of vertical sanitary stacks and storm drain leader lines.
- Each change of direction if the total aggregate change exceeds 90 degrees.
- Main sewer connection outside of building. A 2-way cleanout with dual access plugs (threaded bronze, thermoplastic, or PVC) will be provided at this location. Cleanout will be installed in round cast iron valve box marked “Sewer”.
- Above rim height of all sinks and lavatories on all floors.
- Above rim height of all urinals.
- Up stream of all doubles santees and combinations.

Acceptable Piping Materials

Above Grade Sanitary Waste and Vent Piping: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, neoprene coupler with stainless steel clamp and shield.

Below Grade Sanitary Waste and Vent Piping: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, extra heavy neoprene coupler with stainless steel clamp and shield (Husky, Clamp All, or equal)

Above Grade Vent Piping (Alternate materials)

- Schedule 40 galvanized steel pipe with recessed screwed cast iron drainage fittings.
- Type DWV copper with DWV solder fittings, 50/50 equivalent lead free solder.

Above / Below Grade Waste and Vent Piping (Alternate material for student housing buildings up to 3 stories only) Schedule 40 PVC DWV pipe and DWV fittings with solvent cement joints.

Waste vents shall be located a minimum distance of 30 feet horizontally from HVAC air intakes and occupied roof areas. Additionally, waste vents shall be extended with a support system to a level seven feet above the roof when the roof is occupied (such as in the case of roof mounted greenhouses). Waste vents shall not be located in confined roof wells where HVAC air intakes are located.

Provide accessible cleanouts at all levels of the building the same as required by code for first floor and lower levels.

Avoid sewage lift stations where possible. Gravity flow of sewage is preferred and is usually achievable due to the significant elevation change of most building sites on the campus. If a lift station is required, minimize the required capacity by segregating out upper level waste which can be gravity flow. Lift stations should be duplex pump type, powered by emergency power, with alarm contacts provided as an input to the building management system. Lift stations which handle rest rooms sewage shall be grinder type.

Floor drains shall be equipped with trap primers. Trap primer shall be accessible and equipped with a shut off valve to allow for servicing without system drain down.

22 14 00 - Facility Storm Drainage

Acceptable Piping Materials

Above Grade: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, neoprene coupler with stainless steel clamp and shield.

Below Grade Sanitary Waste: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, extra heavy neoprene coupler with stainless steel clamp and shield (Husky, Clamp All, or equal).

Insulate building storm water piping inside buildings to prevent building damage due to condensation on the pipe exterior.

22 15 00 - General Service Compressed-Air Systems

Acceptable Piping Materials;

Laboratory & Instrument Air Applications: Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing pre-washed, degreased and capped at both ends by manufacture. Fittings pre-washed, degreased and wrapped by the manufacturer. All joints brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to be prevent oxidation.

Alternate material for Shop Air Applications Only: Schedule 40 galvanized steel pipe with treaded 150 pound galvanized malleable iron fittings.

Alternate material for Shop Air Applications Only: Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder.

Building (house) compressed air shall not be shared with the building pneumatic compressed air system for HVAC controls. Building (house) compressed air and HVAC pneumatic control air shall be separate stand-alone systems complete with their own dedicated compressors.

For laboratory and instrument air, provide oil filter with automatic drain, and refrigerated air drier. For shop air applications, confirm requirements with users.

Valving: Compressed air systems shall be provided with manual isolation valves at the following locations:

- At each floor where branch piping connects to a riser.
- At the compressor.
- At branch connections to mains.
- Upstream of any equipment connected to compressed air.

Shop air systems typically require quick couple disconnect fittings for connection of flexible hoses. Confirm requirements with users.

Equipment:

Large house systems: Liquid ring type. (Nash or equal). Provide water saving circulation option with cooling water heat exchanger.

Requirements:
Drain Lines: Daylight to a conspicuous location. Plumb to sewer.

Hammer Arrestors: Install in a maintenance-accessible location for.

22 30 00 - Plumbing Equipment
Domestic Water System
Cold water systems will be sized using CPC flush valve curves, and hot water systems using flush tank curves. Equipment branches and main pipe size will be based on flow requirements without diversity.

All plumbing valves will be located behind a lockable access panels when they are in a concealed location.

Hose bibs will be provided on the roof of each building. The hose bibs will be spaced 50 ft apart, maximum.

Chrome plated stops with gasket seats will be provided for sinks, lavatories, and wash basins when exposed to public view. A hose bibb with vacuum breaker will be provided under the lavatories in each toilet room. Exposed branch water supply piping in toilet rooms and custodial rooms will be chromium plated.

Water hammer arrestors will be provided in the wall, as required, behind a lockable access panel. Trap primers from lavatory tailpieces or water closet flush-o-meter valves for floor drains will also be provided. Isolation valves and unions at equipment connections will be provided.

The water piping will be designed to provide a minimum residual pressure at the most remote water closet of at least 30 psig. Pressure regulators will be installed to comply with California Plumbing Code.

- The maximum water pipe velocity will be 5 ft/sec. for hot water and 8 ft./sec for cold water.
- The minimum supply pipe size will be ½” for one plumbing fixture with a maximum flow of 1.25 gpm.
- Fixture pipe sizes will be as follows: 1-1/2” for a flush valve water closet, ¾” for a shower, ½” for sink and ¾” for hose bibs.

Domestic water piping within the buildings will be “Viega” pro-press or equal copper tubing and shall conform to ASTM B 75 or ASTM B88, Fittings shall be copper in conformance with ASTM B16.18, ASTM B16.22 or ASTM B16.26. O-rings for copper press fittings shall be EPDM.22 32 00 Domestic Water Filtration Equipment

Building Pure Water Systems

Building (house) pure water systems shall be designed to guarantee a water quality level of at least Lab Grade 3 per National Committee for Clinical Laboratory Standards (NCCLS) at all times. Individual laboratories requiring a guarantee of purer water shall install additional purification equipment in the laboratory as part of the building equipment (reference; Letter of Understanding dated 3-19-90, J. West; Physical Plant to K. McCaffrey; Natural Sciences).

The makeup water shall include pre-treatment consisting of a prefilter, a water softener, an activated carbon absorption filter, and a five (5) micron filter followed by treatment which shall consist of a reverse osmosis (RO) unit, mixed bed deionizer exchange service tanks, an ultra-violet light, and a 0.2 micron filter. Make up water capacity shall be determined in conjunction with sizing of the system storage tank.

The entire piping system shall be continually circulated. Dead legs shall be limited to drops to individual lab outlets. (Confirm maximum dead leg distance for each application) The entire circulating volume shall be exposed to ultra-violet light on each circulation. Provide dual stainless steel circulation pumps to allow for continued circulation in the event of pump failure.

The system shall include a storage tank sized to allow make up to occur over a 12 hour period while accommodating maximum anticipated use with 20 % excess capacity. Determination of maximum system use shall include a diversity factors which shall be determined with the University's Representative for each system.

The system shall include a side stream polishing loop which will continually circulate a portion of the water circulating through the system back through the mixed bed deionizers prior to return to the storage tank.

Provide a back pressure valve on the return piping to the storage tank set to assure adequate pressure at the highest most laboratory faucet.

Provide pressure gauges on both sides of all filters and pumps.

Building (house) pure water systems shall be monitored by the Building Management System to provide alarms to the Physical Plant watch stander at the campus Central Heat Plant. Alarms shall be generated in the case of Water quality (conductivity) out of range, low tank level, and no flow.

Acceptable Piping Materials:

Socket fused polypropylene prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system.

Schedule 80 PVC with solvent weld fittings prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system. Stainless steel could be used in some applications.

Confirm piping material with the University's representative on a case by case basis.

Valves shall be diaphragm type.

Valving:

Pure water piping systems shall be provided with manual isolation valves at the following locations:

At each floor where branch piping connects to a riser.

On both sides of piping elements which may need to be removed for servicing including filters, pumps, ultraviolet lights, mixed bed deionizers, and RO Units.

At branch connections to mains.

Provide provisions for system drain down to a floor drain including drain down of storage tanks.

22 35 00 - Domestic Water Heat Exchangers

Tube Bundles: Standard: Cupronickel Alloy. (Note: Standard grade copper tube bundles corrode within 3-5 years.)

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

Equipment: Water Heaters: 10 Year Tank Warranty; State, A.O. Smith, or equal

22 40 00 - Plumbing Fixtures

Lavatory

- Student Housing: Heavy duty commercial grade single handle faucet; deck mounted with 4" centerset, chrome plated finish, vandal resistant handle, brass body, washerless design, rotating ball control mechanism with replaceable nonmetallic seats and stainless steel socket, vandal resistant 0.5 GPM flow restrictors spray outlet, left rod operated metal drain.
  - Preferred Manufacturer: American Standard
    - Lavatory – Wall Hung: American Standard Lucerne Model Numbers:0356.028.020, 20” by 18”, high back.
    - Lavatory – Countertop: American Standard Rondalyn Countertop Sink Model Numbers: 0490.156.020 round, vitreous china, self-rimming front overflow design.

  - Faucets: Preferred Manufacturer and Model: Moen CA83005 Non-mixing Electronic Faucet (0.325 GPM Vandal Resistant Flow). http://www.moen.com/search?search_scope=0&search_terms=CA8305
- Public: Heavy duty commercial grade single handle faucet; deck mounted with 4" centerset, chrome plated finish, vandal resistant handle, brass body, washerless design, rotating ball control mechanism with replaceable nonmetallic seats and stainless steel socket, vandal resistant 0.5 GPM flow restrictor spray outlet, and grid drain.
  - Campus Lavatory Standard (as appropriate):
    - 0.375 gpm Tamper Resistant Laminar Flow Control for Existing Lavatory faucet: Neoperl 40.7059.733
    - Moen 8413 F03 Comm 4” cc .35 GPM Lav Faucet w/o
    - Moen 8551 Sensored Lay Fct
  - Campus Basin Standard: Lavatory Basin with 4” Centerset Holes – American Standard Lucerne 0355.012.020
- Kitchen, Student Housing: Heavy duty commercial grade, single handle faucet, deck mounted with 8" centers, 8" swing spout, chrome plated finish, vandal resistant handle, brass body, rotating stainless steel ball control mechanism with replaceable non-metallic seats and stainless steel socket, vandal resistant 2 GPM flow restrictor aerator.

Shower Valves

- Student Housing: Institutional grade, pressure balancing, adjustable safety limit temperature stop control.
- Public: Institutional grade, pressure balancing, adjustable safety limit temperature stop control.
- Campus Standard: 1.5 GPM Low-Flow Pressure Balancing Surface Mounted Shower System – Hydapipe 1-901

Fixture Traps:
Chrome plated, 17 gauge cast bronze.

Faucets, Supplies, and Trim:
Mechanical Trap Primers:

Standard: Reduce or eliminate use.

Hose Bibbs:

3/4" hose connection, chrome plated, with vacuum breaker, loose key.

Water Meters:
Domestic Water: Nutating disk, magnetic drive, bronze case

Irrigation Water: Turbine type, magnetic drive, bronze case

Pressure Reducing Valve: Bronze with Teflon disk and diaphragm Valves, General Purpose Ball Valves, 2-1/2" and smaller. Provide threaded valve with downstream union.

22 42 13 - Commercial Water Closets, Urinals, and Bidets
Toilets

- Student Apartments:
  - Toilet shall be 1.28 GPF vitreous china round front bowl, with direct flow priming jet and reverse trap. Hydraulic performance exceeding ANSI A112.19.2 requirements.
  - Seat: All plastic, closed front with cover, stainless steel hinge.
  - Campus Preference:
    - 1.28 HET Floor Mount Floor Discharge – American Standard 2234.001.020 1.1 - 1.6
    - 1.28 HET Floor Mount Floor Discharge (ADA) – American Standard 3461.001.020 1.1 - 1.6
- Public Restrooms (including dormitories):
  - Toilets shall be 1.28 GPF, wall hung with support carrier, vitreous china, elongated, flush valve type exceeding ANSI A112.19.2 requirements.
  - Campus Preference:
    - American Standard 2257.101.020 with Afwall 1.1 – 1.6 (1.28)
    - 1.28 GPF HET manual Toilet (No Piston Valve) – Zurn Z6000 AV Model
    - 1.28 GPF HET Sensor Toilet (No Piston) – ZER6000 CPM Medel
  - Flush valve: 1.28 GPF
  - Flush Valves - Toilets (with infrared sensors):

        - Zurn ZER6000AV-ONE-CPM high efficiency flushometer valve
        - Campus no longer installs Flush Valves - Toilets without infrared sensors.
    - Seat: All plastic, open front elongated without cover; stainless steel hinge.
- Alternate toilet for single stall public rest rooms where light use is expected: Toilet shall be 1.28 GPF with hydro-pneumatic pressure assisted flush, vitreous china, floor mounted, elongated bowl, exceeding ANSI A112.19.2 requirements. Seat: All plastic, open front elongated without cover; stainless steel hinge.
- Toilet carrier manufactures Zurn, Josam, Mifab and J.R. Smith
- Not Recommended:
    - Toilet: Zurn Z5615 Series, Zurn Z5655 Series, and Zurn Z5665 Series 1.28 GPF Elongated vitreous china toilet. The toilet bowls are too shallow for proper water scouring when flushed.
    - Toilet Seat: Enlongated Open Front Seat – Bemis 1955SSCT

Urinals

Campus Standard, as appropriate for use/scope:

- Washbrook 0.125 GPF FloWise Washout Top Spud Urinal, or
- *Z5758* "The Retrofit Pint" 0.125 gpf Ultra Low Consumption Urinal System (21" – 24" Footprint)
- Zurn-ZER 6003AV-CPM Model Sensor Operated Battery Powered Flush Valve for ¾" Urinals

Trap Adapters:

Standard: Use when waste line has a tubular trap. Purpose is to allow for easier maintenance.

Preferred Manufacturer and Model: Arrowhead Brass Products, #6001A, 1-1/2" M.I.P. x 1-1/2" Tube Size – Brass Nut, Brass Ferrule with stop. Male in Female UPC-1003.2.
http://www.arrowheadbrass.com/products/

Flushometers:

Flush Valves: Standard: Zurn

22 45 00 - Emergency Plumbing Fixtures

Locations where hazardous materials are used or stored shall be provided with emergency eye washes in compliance with California Code of Regulations, Title 8 (CalOSHA) Section 5162, ANSI Z358.1 - 1990 and applicable ADA requirements.

Emergency eye washes and showers shall be supplied from the potable water system. Industrial water shall not be used to supply these fixtures.

Emergency eye wash and shower control valves shall remain open once activated until intentionally shut off.

Emergency eye washes and showers shall be located a distance no greater than 100 feet from the hazard and shall require no more than 10 seconds for an injured person to reach.

Floor drains shall not be provided at emergency showers in order to contain hazardous materials. The adjacent floor and walls should be designed to provide containment and also to be resistant to water damage. The plumbing design professional should coordinate this issue with the project Executive Architect.

22 47 00 - Drinking Fountains and Water Coolers

Water-Station Water Coolers:

- Hydration Station: Elkay Models LZSTL8WSSP and LZSTL8WSLPSurface Mount bottle filling station. Sanitary, touchless activation with auto 20-second shut-off
- WaterSentry Plus 3000-gallon capacity filtration system, certified to NSF/ANSI 42 & 53 (Lead, Class 1 Partiulate, Chlorine, Taste 7 Odor)
- Integrated silver ion anti-microbial protection
- Fill rate: 1.5 gpm; 1.1 gpm when connected to a remote chiller
- Laminar flow to reduce splash
- Real drain system
- Visual user interface display includes:
- Green Ticker counts bottles saved from waste
- LED visual filter monitor indicates when to replace filter
- Optional remote chiller can be installed within 15 feet of unit to deliver chilled drinking water.

22 62 00 - Vacuum Systems for Laboratory and Healthcare Facilities

Acceptable Piping Materials

Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder for joining 2" and smaller pipe. 15% silver brazing conforming to AWS classification BCuP-5 for joining 2-1/2" and larger piping.

Likely constituents of vacuum exhaust shall be determined and documented with the users to assist determination of vacuum exhaust location and degree of hazard due to discharged gasses.

Vacuum exhaust shall either be piped into the fume exhaust system or directly piped to a safe exterior location 7 feet minimum above the building roof. Vacuum exhaust discharge location should be carefully chosen so as not to create nuisance odors or hazards at operable windows, paths around the building, or other occupied exterior spaces. Vacuum exhausts which may create a nuisance or hazard should be piped to the same point as fume exhaust discharge for entrainment in the fume exhaust plume. House vacuum pump should be electrically interlocked with the fume exhaust fan in cases where the vacuum exhaust is directly connected to the fume exhaust system.

House vacuum pumps shall be protected by a drop out pot on the vacuum side of the unit.

Equipment:
Vacuum Pumps: Large house systems: Liquid ring type. (Nash or equal). Provide water saving circulation option with cooling water heat exchanger.

22 63 00 - Gas Systems for Laboratory and Healthcare Facilities
For Laboratory Buildings with a B Occupancy classification, verify that the quantity of stored gases classified as hazardous materials does not exceed the exempt quantities called out in the Building Code. H occupancy classification may otherwise be required.

Systems and storage for toxic gases shall be in compliance with the Uniform Fire Code including California Amendments in particular Section 8003, covering toxic compressed gases. Additionally, such systems shall be in compliance with the Toxic Gas Ordinance as written by the Western Fire Chiefs Association. Inform both the Executive Architect and University's Representative of requirements for gas storage cabinets, double containment piping, monitoring systems, and maximum quantity limitations in order to assure coordination among all disciplines.

Storage and piping concepts for laboratory gasses should be reviewed with Cal Poly Environmental Health and Safety via the University's Representative early in the design process. Issues include the possibility of asphyxiation (includes inert gasses) in confined spaces, as well as hazardous material concerns.

Laboratory gasses will not be used or stored in rooms having a HVAC system which returns air for recirculation.

Provide seismic restraint systems for all gas canisters and dewars.

Acceptable Piping Materials

Inert, Nontoxic Gases (nitrogen): Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing shall be pre-washed, degreased and capped at both ends by the manufacturer. Fittings shall be pre-washed, degreased and wrapped by the manufacturer. All joints shall be brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to prevent oxidation.

Other Gasses: Piping materials shall be selected on a project by project basis to meet specific program, compatibility, and regulatory requirements. Design for all toxic or hazardous gasses shall be reviewed and approved by EH&S.

Valving:

Laboratory gas systems shall be provided with manual isolation valves at the following locations:

At each floor where branch piping connects to a riser.

At the gas canister connections.

At branch connections to mains.

Upstream of any equipment connected to the laboratory gas system.

22 66 53 - Laboratory Chemical-Waste and Vent Piping

All wet laboratories, all fume hoods with cup sinks, and all other locations using chemicals which could damage standard DWV piping in the event of an accidental spill shall be provided with a lab waste and vent system. The lab waste and vent system shall be made of chemically resistant piping materials. The lab waste and vent system shall merge with the building sewer at a location outside of the building. Prior to combining the two systems, provide a lab waste sampling station (typically a manhole with a sump for sample collection).

Acceptable Piping Materials

Above and below grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant coated cast iron pipe with matching mechanical joint DWV fittings. (Duriron or equal).

Above grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant glass acid piping with glass DWV mechanical joint fittings. (Pyrex , Kymax, or equal).

Above and below grade Lab Waste and Vent Piping

Charlotte Pipe STM F 2618: Standard specification for CPVC Pipe and Fittings for Chemical Waste Drainage. Scope: this specification covers the performance requirements of CPVC pipe, fittings, and solvent cements used in chemical waste drainage systems. F493: Specification for solvent cements for CPVC pipe and fittings. Scope: This specification provides requirements for CPVC solvent cement to be used in joining CPVC pipe and socket fittings.

Lab waste vents shall be located a minimum distance of 50 feet horizontally from HVAC air intakes and occupied roof areas. Additionally, lab waste vents shall be extended with a support system to a level seven feet above the roof when the roof is occupied (such as in the case of roof mounted greenhouses). Lab waste vents shall not be located in confined roof wells where HVAC air intakes are located.

Provide accessible cleanouts at all levels of the building the same as required by code for the first floor and lower levels.

Floor drains shall not be provided in laboratories using hazardous materials as precaution against such materials being discharged to the sewer system. The floor system should be designed to provide containment of such materials in the event of an accidental spill.